Semiconductor Coverage Handbook

House handbook for the Investment Analyst Agent. Prepared by Ife Oluwatayo for the CBS AI Club workshop, November 2026. Version 2. The house views and ranges below are illustrative judgments for a learning project, not investment advice. Current events, prices and filings come from the agent's tools, not from this handbook.

1. Investment rules

1.1 Our thesis

We invest in semiconductor companies that benefit from long-term demand for computing and AI. We want differentiated technology, durable customer demand and strong cash flow, bought at a price that leaves room for a 12% annual return over five years. We are wary of high debt, dependence on a few customers, and growth that relies on a short-lived boom.

1.2 The hurdle and the call

The hurdle is a 12% annual return over five years. The call follows the model's annual return and nothing else: Research if the return is at or above the hurdle, Watch if it is within 3 points below the hurdle, Pass otherwise. The analyst never overrides the call rule. If the analyst disagrees with the result, it changes its assumptions inside the house ranges and explains why, or it says so in the reason.

1.3 What we look for

Differentiated technology: a product customers cannot easily replace, such as a leading accelerator platform, a must-have design tool or a critical manufacturing step. Durable demand: revenue tied to long-term trends in computing, data centers, electrification and industrial automation, not to one product cycle. Strong cash flow: operating margins that hold up through a downturn and capital spending the business can fund itself.

1.4 Debt rule

Read debt from the model as debt divided by operating income. Below 1.0x is comfortable. From 1.0x to 3.0x is acceptable only if the company is paying debt down and its demand is durable; mention it in the reason. Above 3.0x is a leveraged balance sheet: use the low end of the house ranges and say so in the reason. Never calculate the ratio yourself; use the number Return Model returns. For leveraged acquirers (4.4), the ratio is overstated because GAAP operating income is reduced by acquisition amortization (2.3). Still apply the rule, but say in the reason that the ratio is on a GAAP basis.

1.5 Customer concentration rule

Many chip companies sell most of their output to a handful of cloud providers, phone makers or foundries. When the company note says revenue is concentrated in a few customers, do not use the top of the growth range unless the news supports it, and name the concentration risk in the reason.

1.6 What we avoid

We avoid three things. First, peak-cycle margins treated as permanent, because the model holds today's operating margin flat for five years (see 2.4). Second, leveraged acquirers whose growth depends on paying down acquisition debt (see 4.4 and 1.4). Third, stories that depend on one product, one customer or one policy outcome. When a company fits one of these, say which one in the reason and cite this section.

2. How we model

2.1 The model

Return Model takes the latest 10-K numbers and two assumptions from the analyst: average annual revenue growth for five years and the exit multiple of enterprise value to operating income in year five. It holds the current operating margin flat, values the company in year five, adds cash, subtracts long-term debt, divides by shares and compares the year-five price to today's price. The analyst chooses the two assumptions and gives a reason. The code does all the math.

2.2 Choosing growth

Start from the house range in the company note (section 5). Move toward the top of the range only when the news and the company's position support it. Move toward the bottom when the company is near a cycle peak, carries high debt, or depends on a few customers. Growth is a five-year average, so one strong year does not justify the top of the range.

2.3 Choosing the exit multiple

Start from the house range in the company note. The multiple is what the market will pay in year five, when growth has usually slowed, so it should normally be at or below today's multiple. Software-like businesses with recurring revenue earn the highest multiples. Cyclical businesses at peak margins earn the lowest. Acquisition amortization: the model uses GAAP operating income, which for companies that made large acquisitions is reduced by amortization of acquired intangibles. That is an accounting charge, not a cash cost, so these companies trade at higher multiples of GAAP operating income than their peers. The house ranges for these companies (4.4) already allow for this. Do not cut their multiple a second time because it looks high next to peers.

2.4 Peak margins in cyclical businesses

The model holds today's operating margin flat for five years. In a cyclical business at a peak, that overstates future earnings. Memory and equipment companies are the main cases. When the current margin is well above the company's long-run average, use the low end of both house ranges, and say in the reason that the margin is near a cycle peak. Cite this section.

2.5 Data notes and gaps

If Return Model reports a data note, such as missing cash, pre-tax income used in place of operating income, or an operating loss, repeat the note in the reason. If a key input is missing, make the call the model returns but say the result is unreliable.

2.6 Writing the reason

One or two sentences. Name the growth and multiple chosen and why, mention what changed since the last call from Past Calls, and cite the handbook sections used, for example "per 2.4 and 5.5".

3. Industry primer

3.1 The semiconductor value chain

Chips move through four stages. Design software and intellectual property (EDA tools and licensed architectures) are used to design the chip. Chip designers (fabless companies such as Nvidia and Broadcom) design and sell the product. Manufacturers (foundries, memory makers and integrated manufacturers such as Texas Instruments) produce wafers. Equipment makers (such as Applied Materials) sell the machines that make the wafers. Each stage has different margins, cycles and risks.

3.2 The semiconductor cycle

The industry is cyclical. Customers build inventory when chips are scarce and cut orders when they have too much, so revenue and margins swing more than end demand. Memory is the most cyclical segment, equipment follows the capital spending of chipmakers, and analog and design software are the most stable.

3.3 AI and data center demand

Spending on AI data centers drives demand for accelerators, high-bandwidth memory, networking chips and custom chips designed for large cloud providers. This demand is real but concentrated in a few large buyers, and their spending plans can change quickly. Treat AI demand as durable in direction but uncertain in timing.

3.4 Geopolitics and export controls

US export controls limit sales of advanced chips and chipmaking equipment to China, and rules change often. Companies with high China exposure, especially equipment makers, face policy risk that the model cannot capture. Check the news for current rules and mention material exposure in the reason.

4. Company types

4.1 AI compute leaders

Companies whose growth is driven by AI accelerators and data center platforms. Strengths: differentiated technology and very high margins. Risks: concentration in a few cloud customers, competition from custom chips, and margins that may not last. House approach: allow high growth but keep the exit multiple below today's level.

4.2 Memory makers

Companies selling DRAM, NAND and high-bandwidth memory. The product is closer to a commodity and pricing swings with supply. Margins at the top of the cycle are far above the long-run average. House approach: low growth and a low exit multiple, and apply the peak margin rule in 2.4.

4.3 Equipment makers

Companies selling the machines that make chips. Revenue follows chipmakers' capital spending, which is cyclical, and China has been a large market. House approach: moderate growth and a mid-range multiple, lower if margins are near a peak or China exposure is high (see 3.4).

4.4 Leveraged acquirers

Companies that have used debt to buy large businesses. They can generate strong cash flow, but much of it goes to repaying debt, and integration can disappoint. House approach: use the low end of the growth range until debt divided by operating income falls toward 1.0x, and apply the debt rule in 1.4. Their multiple ranges are set on a GAAP basis that includes acquisition amortization (2.3).

4.5 Design software and IP

Companies selling the software and licenses used to design chips. Revenue is recurring, customers rarely switch, and margins are stable through the cycle. House approach: steady growth and the highest multiples in our coverage, adjusted for any acquisition debt.

4.6 Analog and industrial chips

Companies selling power management and signal chips for industrial, automotive and consumer markets. Products last for years and customers are spread out, but growth is modest and heavy factory spending can depress cash flow for a period. House approach: low to moderate growth and a mid-range multiple.

5. Company notes

5.1 NVIDIA (NVDA)

Type: AI compute leader (4.1). The leading supplier of AI accelerators and the software platform built around them, with data center revenue the large majority of sales. Fit with thesis: strong on differentiated technology and cash flow. Risks: a small number of cloud customers account for a large share of revenue (1.5), large customers are designing their own chips, and current margins are at historic highs (2.4). House growth range: 12% to 22%. House exit multiple range: 18x to 28x. Start in the middle of both ranges. Move toward the low end if the news shows customer concentration rising or custom chips taking share, and toward the high end if demand is broadening beyond the largest cloud buyers.

5.2 Broadcom (AVGO)

Type: AI networking and custom chips, and a leveraged acquirer (4.4). Sells networking chips and custom AI accelerators to large cloud providers, and infrastructure software after its acquisition of VMware, which it funded largely with debt. Fit with thesis: differentiated technology and strong cash flow, but debt and customer concentration both apply. Risks: debt (1.4), reliance on a few custom chip customers (1.5). House growth range: 10% to 18%. House exit multiple range: 22x to 32x, set on a GAAP basis that includes acquisition amortization (2.3). Apply the debt rule and cite 1.4, 2.3 and 4.4.

5.3 Applied Materials (AMAT)

Type: equipment maker (4.3). The largest supplier of chipmaking equipment across deposition, etching and inspection. Fit with thesis: benefits whichever chip designer wins, and has strong cash flow. Risks:

cyclical demand tied to chipmakers' capital spending, and meaningful China exposure subject to export controls (3.4). House growth range: 5% to 10%. House exit multiple range: 14x to 20x. Use the low end if the news points to slowing capital spending or tighter export rules.

5.4 Synopsys (SNPS)

Type: design software and IP (4.5), and a leveraged acquirer (4.4) after its acquisition of Ansys, funded partly with debt. One of two leading suppliers of chip design software. Fit with thesis: very strong on differentiated technology and recurring demand. Risks: acquisition debt and integration (1.4), and China exposure. House growth range: 8% to 13%. House exit multiple range: 30x to 40x, set on a GAAP basis that includes acquisition amortization (2.3). Apply the debt rule and cite 1.4, 2.3 and 4.5.

5.5 Micron Technology (MU)

Type: memory maker (4.2). Sells DRAM, NAND and high-bandwidth memory for AI servers. Fit with thesis: benefits from AI demand, but memory is the most cyclical part of the industry, which is the short-lived boom our thesis warns about (1.6). Risks: peak-cycle margins treated as permanent (2.4), and new supply bringing prices down. House growth range: 0% to 8%. House exit multiple range: 6x to 10x. When the current operating margin is far above its long-run average, use the low end of both ranges and cite 2.4 and 4.2.

5.6 Texas Instruments (TXN)

Type: analog and industrial chips (4.6). Sells tens of thousands of analog and embedded products to a broad customer base, and manufactures most of them in its own factories. Fit with thesis: durable, diversified demand and a long record of returning cash to shareholders. Risks: modest growth, sensitivity to the industrial and automotive cycle, and heavy factory investment that has weighed on free cash flow. House growth range: 4% to 8%. House exit multiple range: 16x to 22x.